

EEET202

PHINES & TRANSFORMERS

MODULE: 4

deal Transformer, Dot Convention
and KVA Rating

Prepared By,

ADARSH SR

Asst. Professor

EEE Department

SCET Kodakara



Syllabus

ers –constructional details, principle of operation, EMF equation, convention, magnetising current, transformation ratio, phasor no load and on load, equivalent circuit, percentage and per unit regulation. Transformer losses and efficiency, condition for 7/A rating. Testing of transformers– polarity test, open circuit test, Sumpner's test – separation of losses, all day efficiency.Parallel transformers– numerical problems

Ideal Transformer

Transformer is an imaginary transformer which has

- No losses (no winding resistance)

- No loss in core

- No leakage flux

Transformer gives output power exactly equal to the

Input power of an ideal transformer is 100%

Ideal Transformer

Resistance: Resistance of primary as well as secondary of an ideal transformer is zero.

Permeability of the core: Higher the permeability, lesser reluctance for flux establishment i.e; less magnetizing current is required to magnetize the transformer core.

All flux is confined to core & the windings

An ideal transformer does not have any losses such as copper loss, eddy current loss etc.

Dot Convention

The phase relation between primary and secondary voltages depends on how each winding is wrapped around the core. In fig (1), primary and secondary windings of both transformers are wrapped around the core in the same direction.

In fig (2), primary winding is wound around the core in the opposite direction to the secondary winding. The voltage induced in the secondary winding in fig (2) is opposite to the voltage induced in the secondary winding in fig (1) as compared with the induced voltage in secondary winding in fig (1). The direction of secondary current (I_s) is opposite from the direction of secondary current (I_s) in fig (1).

Primary voltage and current are in phase in fig (1).

Primary voltage and current are 180° out of phase in fig (2).

Dot Convention

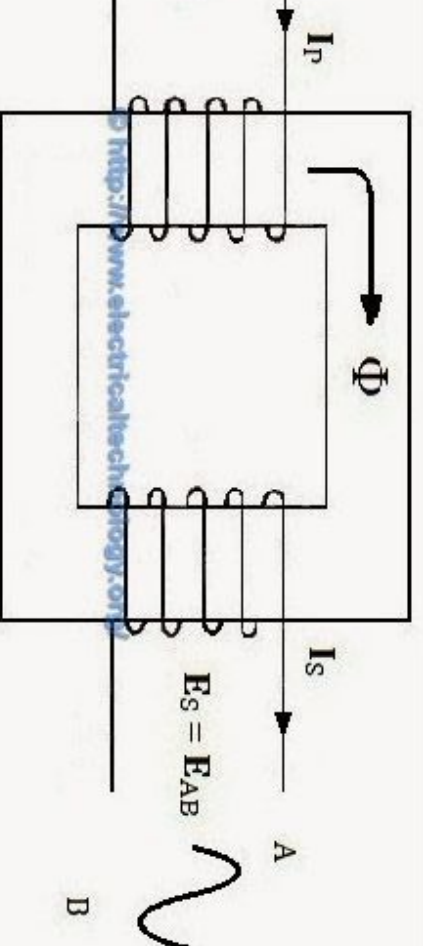


Fig (1)

Primary and Secondary voltage in Phase

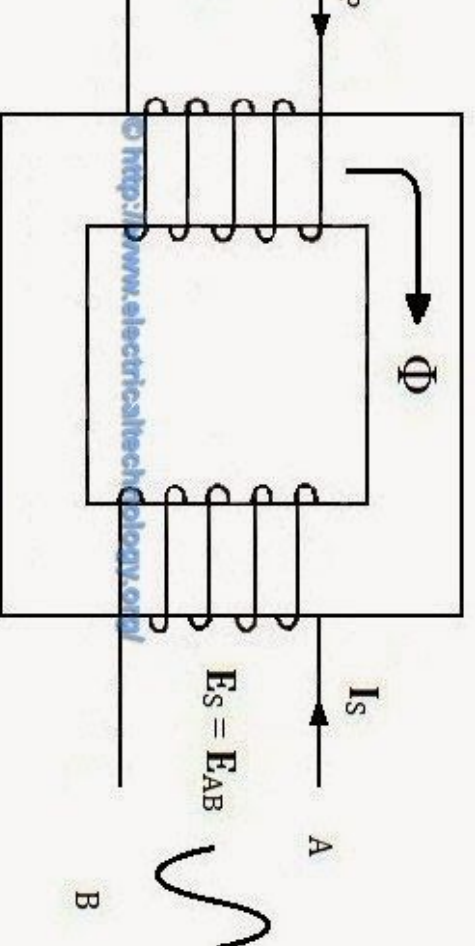


Fig (2)

Primary and Secondary voltage 180° out of Phase

Dot Convention

on the top of primary and secondary terminals as
d (4).

placed at the top in both primary and secondary

primary and secondary current and voltages are in

primary and secondary voltages (V_p and V_s) have similar
primary and secondary (I_p and I_s) currents are same

Dot Convention

is positioned at the top in primary terminal and the placed at bottom of secondary terminal

primary and secondary current and voltages are 180°

primary and secondary voltages (V_p and V_s) sine e to each other

and secondary currents (I_p and I_s) are opposite in

Dot Convention

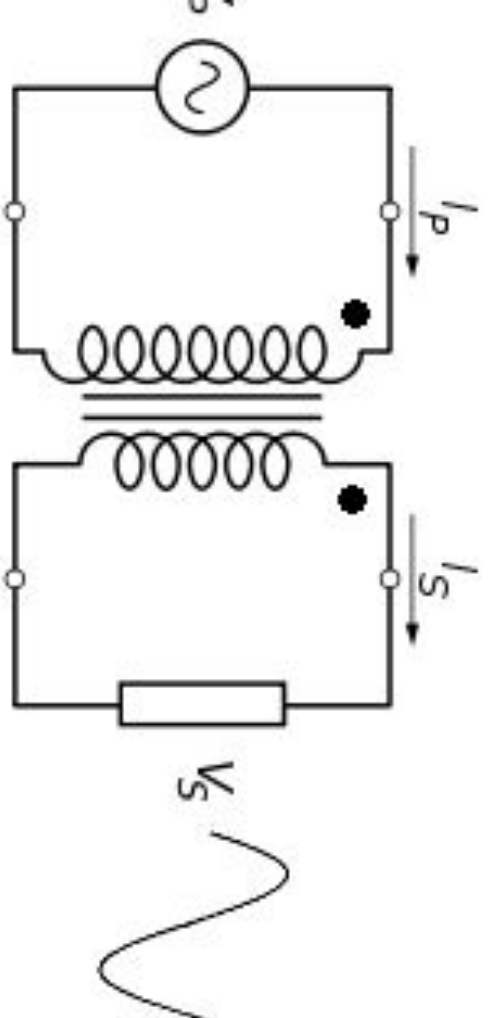


Fig (3)

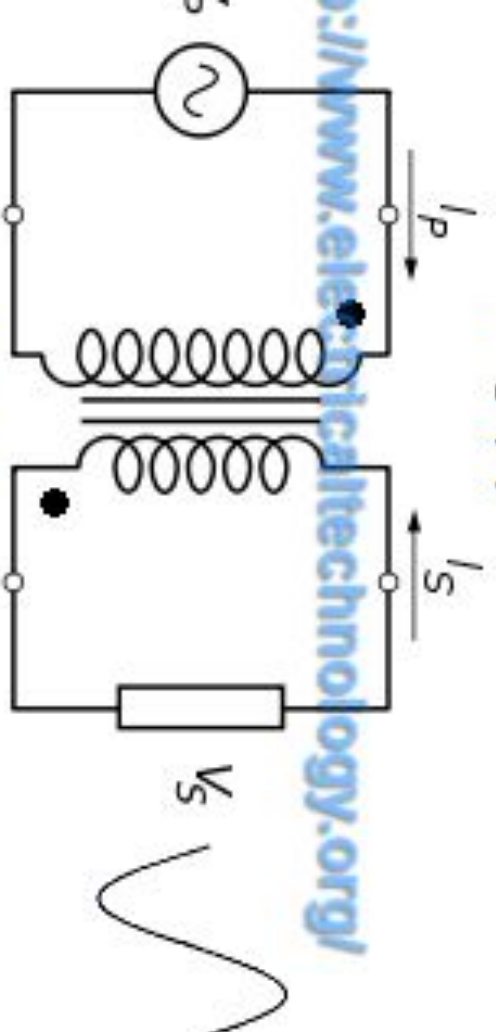


Fig (4)

Rating of Transformer

I^2R) in the transformer depends on the current 'I' and the resistance 'R'.
e.g.

Loss depends on the voltage 'V' as the frequency of the supply is constant.

Losses depend on the power factor ($\cos \phi$) of the load.

Temperature rise & hence the rating of the transformer

depends on V & I only, the transformer rating is specified as VA or KVA rating & called VA rating or KVA rating.

Rating of Transformer

Primary VA rating remains same
usually expressed in KVA

$$\frac{V_2}{V_1} = \frac{I_1}{I_2} = K$$

$$V_1 I_1 = V_2 I_2$$

$$\text{transformer} = \frac{V_1 I_1}{1000} = \frac{V_2 I_2}{1000}$$

Rating of Transformer

primary & secondary currents indicate the safe limits of currents which transformer windings can

$$I_1 \text{ load} = \frac{\text{KVA rating} * 1000}{V_1}$$

$$I_2 \text{ load} = \frac{\text{KVA rating} * 1000}{V_2}$$